

Lab 4

NLI for insights and Databases

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1. Objective

The main goal of this lab was to see how natural language can be used to interact with a database and get useful insights without writing complex SQL queries.

Using the Netflix titles dataset, we built a simple Natural Language Interface (NLI) that understands a few types of questions and returns answers directly from the database.

Through this process, we also compared two different database approaches—SQLite and InfluxDB—to understand their strengths and limitations for data analysis tasks.

2. Database Choice

For our implementation, we decided to use SQLite as the main database and treat InfluxDB as an optional comparison.

SQLite was a practical choice because it runs natively in Colab, requires no setup, and works very well with pandas for quick data analysis. It also fits the nature of our dataset, which is mostly categorical and not time-based.

InfluxDB, on the other hand, is more powerful for time-series or streaming data.

We reviewed how it could be connected through an API client, but given our static dataset, the benefits did not outweigh the extra configuration steps.

Overall, SQLite allowed us to focus more on the logic of NLI and less on infrastructure setup.

3. Q1 — Total Number of Titles

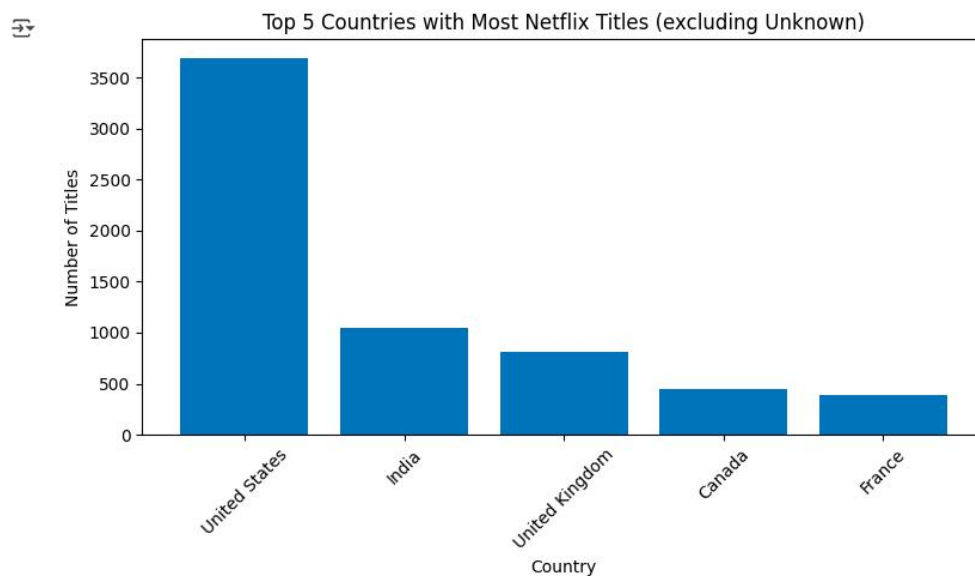
After loading the dataset `netflix_titles.csv` into SQLite, we verified that the table contains **8,807 records**.

This matches the Kaggle dataset version and confirms that the data was loaded and stored correctly in the database.

4. Q2 – Top 5 Countries with Most Titles

Using SQL queries and pandas explosion on the comma-separated country field, the five countries with the largest Netflix libraries were identified as:

Rank	Country	Titles
1	United States	3,690
2	India	1,046
3	United Kingdom	806
4	Canada	445
5	France	393



Note: Unknown country (missing) = 831 (9.4% of 8807)

Approximately **831 records (9.4 % of 8,807)** had no country information.

These were excluded from the ranking but reported separately to keep the statistics transparent.

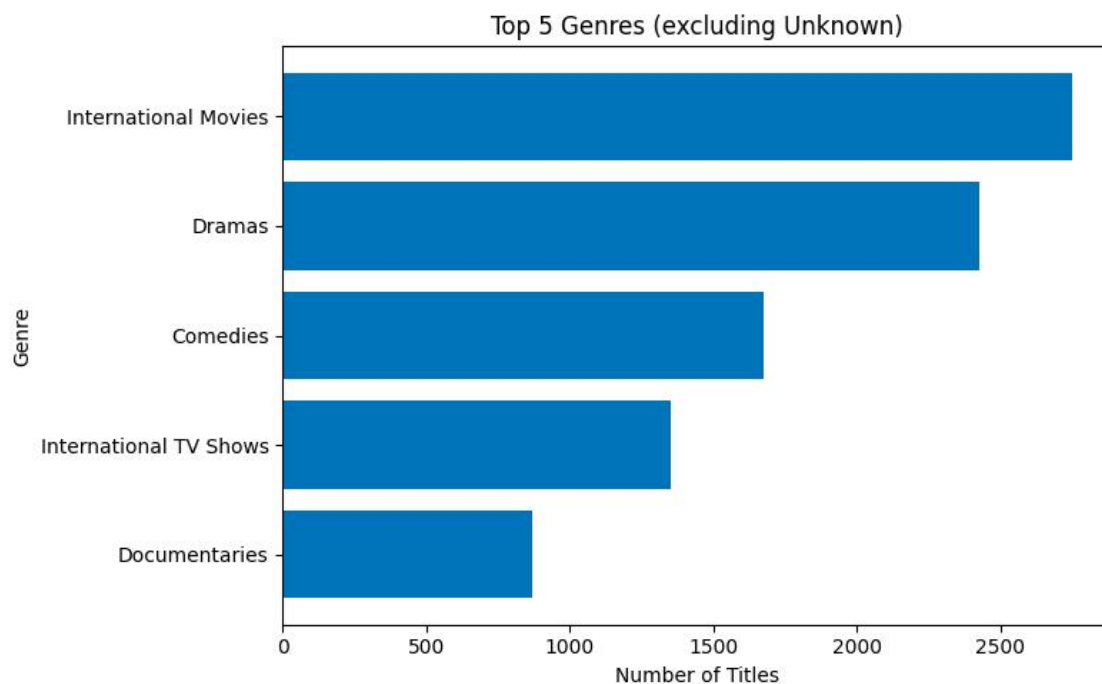
The United States clearly dominates Netflix's catalog, followed by India and the United Kingdom.

This distribution reflects Netflix's global expansion and its strong partnerships with production companies in these regions.

5. Q3 – Most Common Genres

The listed_in column was split and exploded to count individual genres. The top five genres are:

Genre	Titles
International Movies	2,752
Dramas	2,427
Comedies	1,674
International TV Shows	1,351
Documentaries	869



These results suggest that Netflix focuses heavily on story-driven and globally appealing content.

The dominance of “International Movies” and “Dramas” also shows how Netflix positions itself as a platform for cross-cultural storytelling.

6. Q4 – NLI (Natural Language Interface) Queries

A lightweight NLI module was implemented to map natural-language questions to SQL queries. Examples:

User Question	Database Query	Result (Excerpt)
“total shows”	SELECT COUNT(*) FROM netflix	8,807
“top countries”	Pre-computed pandas aggregation	US > India > UK > Canada
“top genres”	Pre-computed pandas aggregation	Intl Movies > Dramas > Comedies
“recent shows (2020 +)”	SELECT title, release_year FROM netflix WHERE release_year >= 2020	Blood & Water (2021), Ganglands (2021), Kota Factory (2021)...

Each query returned the correct information directly from SQLite.

This demonstrates that even a simple keyword-based approach can make database interaction more intuitive and user-friendly.

```

Test 1: Total number of shows
total
8807

---

Test 2: Top countries (excluding Unknown)
country total
United States 3690
India 1046
United Kingdom 806
Canada 445
France 393

---

Test 3: Top genres (excluding Unknown)
genre total
International Movies 2752
Dramas 2427
Comedies 1674
International TV Shows 1351
Documentaries 869

---

Test 4: Recent shows (2020 and newer)
title release_year
Blood & Water 2021
Ganglands 2021
Jailbirds New Orleans 2021
Kota Factory 2021
Midnight Mass 2021

[✓] NLI test complete.

```

7. Q5 — Comparison: SQLite vs InfluxDB 3

Aspect	SQLite (Relational)	InfluxDB 3 (Time-Series / NoSQL)
Setup	Built-in, no configuration needed in Colab	Requires external server / token / bucket
Query Language	SQL — ideal for grouping, filtering	Flux / SQL+ — optimized for time-based queries
Data Model	Tabular (rows & columns)	Measurement-based (time, tags, fields)
Performance	Excellent for small datasets	Better for high-volume time-series data
Ease of Use	Easiest to integrate with Python and pandas	Requires API client setup and schema design
Best Use Case	Static categorical data (Netflix metadata)	Dynamic data (e.g., content additions over time)

SQLite is the most appropriate choice for this assignment, providing simple SQL querying and smooth integration with Colab and pandas. InfluxDB would be preferable for time-series or real-time datasets but adds unnecessary overhead here.

8. Summary and Insights

- The dataset contains 8,807 titles from multiple countries and genres.
- The United States leads production volume, followed by India, the U.K., Canada, and France.
- Around 9 % of records have missing country data, which were handled transparently.
- The most frequent genres are International Movies and Dramas, reflecting Netflix's global content strategy.
- The NLI system makes querying straightforward, enabling users to extract insights in natural language rather than SQL.
- SQLite proved to be lightweight, fast, and ideal for this structured dataset, while InfluxDB would be more suitable for time-series or streaming data.